

Xinwei Qiang

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Research interests: GPU systems, compilers, and distributed systems.

Publications

- [*NeurIPS* '26] AsymHP: Load-Balanced Sparse Attention for Video Diffusion Transformers.
Xinwei Qiang*, Yue Guan*, Ruihan Zhu, Mihir Jagtap, Zaifeng Pan, Zhongkai Yu, Chang Chen, Zhengding Hu, Yufei Ding, Adnan Aziz
- [*ASPLOS* '27] Impeller: Scalable and Efficient Zero-Knowledge Proof Computation Framework via Graph Compilation.
Liukun Yu*, **Xinwei Qiang***, Shixuan Sun, Chen Chen, Jingwen Leng, Jin Dong
- [*arXiv* '26] Beyond Parallel Blindness: Information Floors and Model Gaps in Block Drafting.
Xinwei Qiang, Xiang Fang, Chang Chen, Zaifeng Pan, Yue Guan, Yufei Ding
- [*SC* '26] FluxZK: Scalable and Efficient Zero-Knowledge Proof Computation via GPU Acceleration.
Xinwei Qiang*, Liukun Yu*, Xiyu Wang, Zhengyi Li, Shixuan Sun, Jingwen Leng, Chen Chen, Jiaping Gui, Zhenzhe Zheng, Jin Dong, Minyi Guo
- [*arXiv* '26] GF-DiT: Scheduling Parallelism for Diffusion Transformer Serving.
Xinwei Qiang, Yifan Hu, Shixuan Sun, Jing Yang, Han Zhao, Chen Chen, Yu Feng, Jingwen Leng, Minyi Guo
- [*ICML* '26] TritonGym: A Benchmark for Agentic LLM Workflows in Triton GPU Code.
Yue Guan*, Yichen Lin*, Xu Zhao, Jianzhu Yao, **Xinwei Qiang**, Zhongkai Yu, Pramod Viswanath, Yufei Ding, Adnan Aziz
- [*OSDI* '26] Syncopate: Efficient Multi-GPU AI Kernels via Automatic Chunk-Centric Compute-Communication Overlap.
Xinwei Qiang*, Yue Guan*, Zhengding Hu, Keren Zhou, Yufei Ding, Adnan Aziz
- [*ICLR* '26] DASH: Deterministic Attention Scheduling for High-throughput Reproducible LLM Training.
Xinwei Qiang, Hongmin Chen, Shixuan Sun, Jingwen Leng, Xin Liu, Minyi Guo
- [*SOSP* '25] Mercury: Unlocking Multi-GPU Operator Optimization for LLMs via Remote Memory Scheduling.
Yue Guan, **Xinwei Qiang**, Zaifeng Pan, Daniels Johnson, Yuanwei Fang, Keren Zhou, Yuke Wang, Wanlu Li, Yufei Ding, Adnan Aziz

* Equal contribution.

Research Appointments

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|---|---------------------|
| Research Assistant , EPCC Lab, Shanghai Jiao Tong University
<i>Co-advised by Profs. Jingwen Leng and Shixuan Sun</i> | Jul 2024 – Jun 2026 |
| Research Intern , Picasso Lab, University of California San Diego
<i>Advisor: Prof. Yufei Ding</i> | Jul 2025 – Dec 2025 |

Education

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| University of California San Diego — Ph.D. in Computer Science
<i>Advisor: Prof. Yufei Ding</i> | Sept 2026 – Present |
| Shanghai Jiao Tong University — B.E. in Computer Science
John Hopcroft Class, an honors program in computer science. | Sept 2022 – Jun 2026 |

Technical Skills

Programming: C/C++, CUDA, Python, Rust, Triton